

Retrieval-augmented generation often suffers from hallucinations and misattributed citations where generated text points to incorrect document pages or fabricated statements.

Verbatim text span alignment maps retrieved chunk elements directly to PDF selectable text layer character spans, eliminating arbitrary percentage overlays and fake bounding boxes.

Hybrid retrieval combines dense vector embeddings with PostgreSQL tsvector full-text search, merged through reciprocal rank fusion with parameter  $k=60$ .

Multilingual cross-encoder reranking re-evaluates top 40 candidate chunks to distill the top 6 evidence items with minimal positional bias.

Bounding box origin normalization converts PDF bottom-left points into standard top-left DOM coordinates for synchronized PDF.js viewer rendering.

Claim-to-citation validation checks semantic support between the generated assertion and the referenced source span, demoting unsupported claims to unresolved status.